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$f(x,y) = x^2 + 2xy + 3y^2$ in $a(2,1)$ in de richting $h(1,-1)$

$$\lim_{\lambda \to 0} \frac{f(a + x\lambda, b + 4\lambda) - f(a,b)}{\lambda}$$

$a(a,b) = a(2,1)$

$h(x,y) = h(1,-1)$

$$= \lim_{\lambda \to 0} \frac{f(2 + 1\lambda, 1 - 1\lambda) - f(2,1)}{\lambda}$$

$$= \lim_{\lambda \to 0} \frac{f(2 + 1\lambda, 1 - 1\lambda) - f(2,1)}{\lambda}$$

$$= \lim_{\lambda \to 0} \frac{(2+\lambda)^2 + 2(2+\lambda)(1-\lambda) + 3(1-\lambda)^2 - (4+4+3)}{\lambda}$$

$$= \lim_{\lambda \to 0} \frac{(4 + 4\lambda + \lambda^2) + (4 - 4\lambda + 2\lambda - 2\lambda^2) + (3 - 6\lambda + 3\lambda^2)}{\lambda}$$

$$= \lim_{\lambda \to 0} \frac{2\lambda^2 - 4\lambda}{\lambda}$$

$$= \lim_{\lambda \to 0} (2\lambda - 4) = -4$$

References